Annwoy Roy Choudhury

Master's Thesis Research Scholar
Institute for Quantum Optics and Quantum Information (IQOQI) & University of Innsbruck (UIBK), Innsbruck, Austria
Fifth-Year Integrated BS-MS Dual Degree Student (equivalent to BSc + MSc)
Department of Physical Sciences, Indian Institute of Science Education and Research (IISER), Mohali, India

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Career Objectives

To pursue a career in Quantum Physics, specializing in Quantum Information, Quantum Many-Body Physics, and Tensor Networks. My goal is to develop mathematical and computational frameworks that bridge quantum information with quantum field theory, relativity, and the foundations of quantum mechanics. I am immensely curious to keep learning and rigorously solving fundamental problems in physics, with a focus on understanding entanglement in complex quantum systems.

Research Experience

Master's Thesis Research Scholar - Institut für Quantenoptik und Quanteninformation (IQOQI) Innsbruck, Austria
Institut für Theoretische Physik | ***Supervisor: Prof. Dr. Klemens Hammerer*** *Jun 2026 – Present*

- Simulating highly entangled, non-Gaussian quantum states for sub-SQL metrological precision using Matrix Product States (MPS) and Matrix Product Operators (MPO), analyzing One-Axis Twisting (OAT) and power-law Ising Hamiltonians. [GitHub Repository]
- Developing novel open-system tensor network architectures for quantum metrology applications, employing Bañuls local-equilibration and Tensor Jump Methods (TJM) to capture dynamics under local T_1 relaxation and T_2 dephasing.
- Analyzing the generalized Ramsey echo landscape, demonstrating Heisenberg-like scaling for Over-Un-Twisting (OUT) protocols and mapping algorithmic breakdown floors at large system sizes (N=200).
- Developing a QFI-preserving compression framework for LPTN tensor networks, with rigorous constraints on metrologically active moments and provable positivity preservation.

Nordita Summer Fellow - The Nordic Institute for Theoretical Physics (NORDITA) Stockholm, Sweden
Department of Physics, Stockholm University (remote)
Supervisors: Dr. Sofia Qvarfort, Dr. Jack Clarke *Aug 2025 – Present*

- Developed a displaced-equilibrium protocol to bypass the Rotating-Wave Approximation (RWA) parity filter, promoting odd Taylor coefficients into the measurable even sector via binomial mixing.
- Derived bounds for optimal probe states, proving that estimating K parameters with an invertible Quantum Fisher Information (QFI) matrix fundamentally requires at least $K+1$ Fock components.
- Demonstrated Heisenberg-like scaling (N^6) for a minimal three-component optimal probe state under a two-parameter Newtonian gravity model, establishing the flat geometry of the model-independent parameter manifold.

Research Scholar - Zentrum für Optische Quantentechnologien, Universität Hamburg (UHH) Hamburg, Germany
Institute for Quantum Physics | ***Supervisor: Prof. Dr. Peter Schmelcher*** *May 2025 – Nov 2025*

- Developed a modular Python framework from scratch for MPS functionalities, tensor decompositions (SVD, QR, RQ), and QAOA simulation. [GitHub Repository]
- Analyzed exact vs. compressed simulation strategies, demonstrating that global Variational compression consistently outperforms greedy SVD methods.
- Investigated infidelity scaling across different graph topologies (Linear Chain vs. Complete Graph), identifying fidelity-floor limits and validating the Multiplicative Fidelity Law for estimating inexact overlaps.

Research Scholar - Max-Planck-Institut für Kernphysik (MPIK) Heidelberg, Germany
Quantum Dynamics and Control Division | ***Supervisor: Prof. Dr. Thomas Pfeifer*** *May 2024 – Sep 2024*

- Leveraged `QuTiP` to solve the time-dependent Schrödinger equation to simulate optical density spectra for Hydrogen atoms interacting with short laser pulses encoding handwritten digits.
- Utilized Fourier transformations to relate dipole responses to optical density, analyzing transition probabilities and population dynamics via Fano lineshapes.

Summer Intern - Indian Statistical Institute (ISI) Kolkata, India
Physics and Applied Mathematics Unit | ***Supervisor: Dr. Ramij Rahaman*** *Jun 2023 – Jul 2023*

- Explored fundamental literature in Quantum Circuits and systematically solved algorithmic problems in Quantum Error Correcting Codes.

Publications & Preprints

- **A. Roy Choudhury**, J. Clarke, S. Qvarfort. *"Quantum reconstruction of arbitrary potentials with a displaced mechanical oscillator."* Manuscript in preparation (2026).
- **A. Roy Choudhury**, J. Clarke, S. Qvarfort. *"Seeing odd forces with a quantum oscillator: displaced sensing and the multiparameter bound."* Manuscript in preparation (2026).

Education

Universität Innsbruck (UIBK) — Innsbruck, Tirol, Austria
Master's Thesis Research, Theoretical Physics — *Jun 2026 – Present*

- Long-term research conducted as part of the integrated BS-MS program of IISER Mohali.

Indian Institute of Science Education and Research (IISER) — Mohali, Punjab, India
Integrated BS-MS Dual Degree in Physics (five-year program, equivalent to BSc + MSc)
Minor: Quantum Information Science — *Nov 2022 – Present*

- **Cumulative Point Index (CPI):** 9.0 out of 10

Bidhannagar Government High School — Salt Lake, West Bengal, India
Higher Secondary (12th Grade) | *Percentile: 99.66* | *Marks: 93.00%* — *Jun 2019 – May 2021*

Selected Talks and Presentations

Reconstructing Unknown Potential with a Quantum Oscillator — *Universität Innsbruck · Oct 2026*

- Discussed my work as a Nordita Fellow on the reconstruction of arbitrary unknown potential, how to bypass the parity filter imposed by Rotating Wave Approximation and how to reach Heisenberg-like scaling

Dynamic Critical Phenomena: The Hohenberg-Halperin Classification — *IISER Mohali · Apr 2026*

- Analyzed the Hohenberg-Halperin classification and compared stochastic Langevin equations (Models A and B) to describe critical slowing down near phase transitions.

Rigorous Classification of 1D Gapped Quantum Phases via Matrix Product States — *IISER Mohali · Apr 2026*

- Mapped the transition from Landau Symmetry Breaking to Symmetry-Protected Topological (SPT) Order using group cohomology and Matrix Product States.

Fidelity Limits in Matrix Product State Simulation of Max-Cut QAOA [Poster] — *QFTA Conference · Dec 2025*

- Analyzed infidelity and computational limits in simulating QAOA circuits via Matrix Product States.

Classical Simulation of Quantum Circuits using Tensor Networks — *Universität Hamburg · Jul 2025*

- Presented methodologies for classically simulating QAOA and evaluated the performance of tensor network compression algorithms.

On the Reality of Quantum State: PBR Theorem — *IISER Mohali · Apr 2025*

- Discussed the PBR theorem's implications for the ontological status of quantum states.

Entanglement and Quantum Phase Transitions — *IISER Mohali · Nov 2024*

- Discussed tensor networks and quantum information tools to analyze quantum phase transitions.

What the Heck is Quantum Mechanics? — *IISER Mohali · Nov 2023*

- Delivered an introductory presentation on the foundational postulates and phenomena of quantum mechanics.

Scholastic Achievements

- **ICTS-TIFR Long Term Visiting Students Program Fellow:** Selected to work with Prof. Manas Kulkarni at the International Centre for Theoretical Sciences (ICTS-TIFR), Bengaluru, India [April 2026]. (Offer declined)
- **IBM Quantum Excellence Badge Awardee:** Qiskit Global Summer School (Covering modern quantum computing, error mitigation, and simulating molecular Hamiltonians with SQD) [July 2025].
- **IASc-INSA-NASI Summer Research Fellow:** Awarded a two-month research fellowship to work with Prof. M. Senthilvelan at Bharathidasan University, Tiruchirapalli, India [April 2024]. (Offer declined)
- **Regional Level Awardee:** Madhava Mathematics Competition organised by HBCSE, T.I.F.R, Mumbai [April 2023].
- **First Place for Outstanding Achievement Awardee:** American Mathematical Competitions 12 (AMC 12) [April 2020].
- **SAT India Top Performer Awardee:** 99th Percentile. Mathematics: 800/800 [Dec 2020].

Technical Skills & Relevant Coursework

- **Programming:** Python (Advanced), TeNPy, QuTiP, Qiskit, C, NumPy, SciPy (incl. `scipy.optimize`, `scipy.linalg`), mpi4py, h5py, matplotlib, LaTeX, Git.
- **Physics Courses:** Quantum Phases of Matter and Phase Transitions, Statistical Physics of Fields, Quantum Computation and Quantum Information, Open Quantum Systems, Foundations of Quantum Mechanics, Implementations of Quantum Information Processing, Solid State Physics, Atomic and Molecular Physics, Nuclear Physics, Modelling Complex Systems, Classical Mechanics (2 Courses), Quantum Mechanics (2 Courses), Electromagnetism (2 Courses), Waves and Optics, Thermodynamics and Statistical Mechanics (2 Courses).
- **Mathematics Courses:** Real Analysis, Group Theory, Linear Algebra, Differential Geometry, Statistics and Probability, Mathematical Methods for Physicists.

Academic Activities, Extracurricular Activities & Languages

- **Teaching Assistant:** Electromagnetism, IISER Mohali.
- **Pedagogical Outreach:** Creating a video lecture series on Differential Geometry.
- **Editorial Leadership:** Senior Editor for *Manthan Magazine*, IISER Mohali.
- **Scientific Communication:** Presented research work to the public at Open House Day, MPIK, Heidelberg.
- **Photography:** Wildlife, astro, street and travel photography with acceptances in various National and International Photography Salons; volunteered as Official Photographer in various events.
- **Birding:** Certified birdwatcher and bird population counter.
- **Music & Literature:** First Place in State Level Rabindra Sangeet Competition; published original Bengali and English literature, performed as a vocalist, and directed self-written musical scripts.
- **Languages:** Bengali (Native), English (Full Professional), German (Elementary), Spanish (Elementary).

References

- **Prof. Dr. Klemens Hammerer**
  Institute for Theoretical Physics, University of Innsbruck
  [klemens.hammerer@uibk.ac.at]
- **Dr. Sofia Qvarfort**
  Department of Physics, Stockholm University & NORDITA
  [sofia.qvarfort@fysik.su.se]
- **Prof. Dr. Thomas Pfeifer**
  Quantum Dynamics and Control Division, Max-Planck-Institut für Kernphysik, Heidelberg
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